

4.7 Acid/base equilibria

Students will be assessed on their ability to:

- a demonstrate an understanding that the theory about acidity developed in the 19th and 20th centuries from a substance with a sour taste to a substance which produces an excess of hydrogen ions in solution (Arrhenius theory) to the Brønsted-Lowry theory
- b demonstrate an understanding that a Brønsted-Lowry acid is a proton donor and a base a proton acceptor and that acid-base equilibria involve transfer of protons
- c demonstrate an understanding of the Brønsted-Lowry theory of acid-base behaviour, and use it to identify conjugate acid-base pairs
- d define the terms pH, K_a and K_w , pK_a and pK_w , and be able to carry out calculations relating the pH of strong acids and bases to their concentrations in mol dm^{-3}
- e demonstrate an understanding that weak acids and bases are only slightly dissociated in aqueous solution, and apply the equilibrium law to deduce the expressions for the equilibrium constants K_a and K_w
- f analyse the results obtained from the following experiments:
 - i measuring the pH of a variety of substances, e.g. equimolar solutions of strong and weak acids, strong and weak bases and salts
 - ii comparing the pH of a strong acid and a weak acid after dilution 10, 100 and 1000 times
- g analyse and evaluate the results obtained from experiments to determine K_a for a weak acid by measuring the pH of a solution containing a known mass of acid, and discuss the assumptions made in this calculation
- h calculate the pH of a solution of a weak acid based on data for concentration and K_a , and discuss the assumptions made in this calculation
- i measure the pH change during titrations and draw titration curves using different combinations of strong and weak monobasic acids and bases
- j use data about indicators, together with titration curves, to select a suitable indicator and the use of titrations in analysis